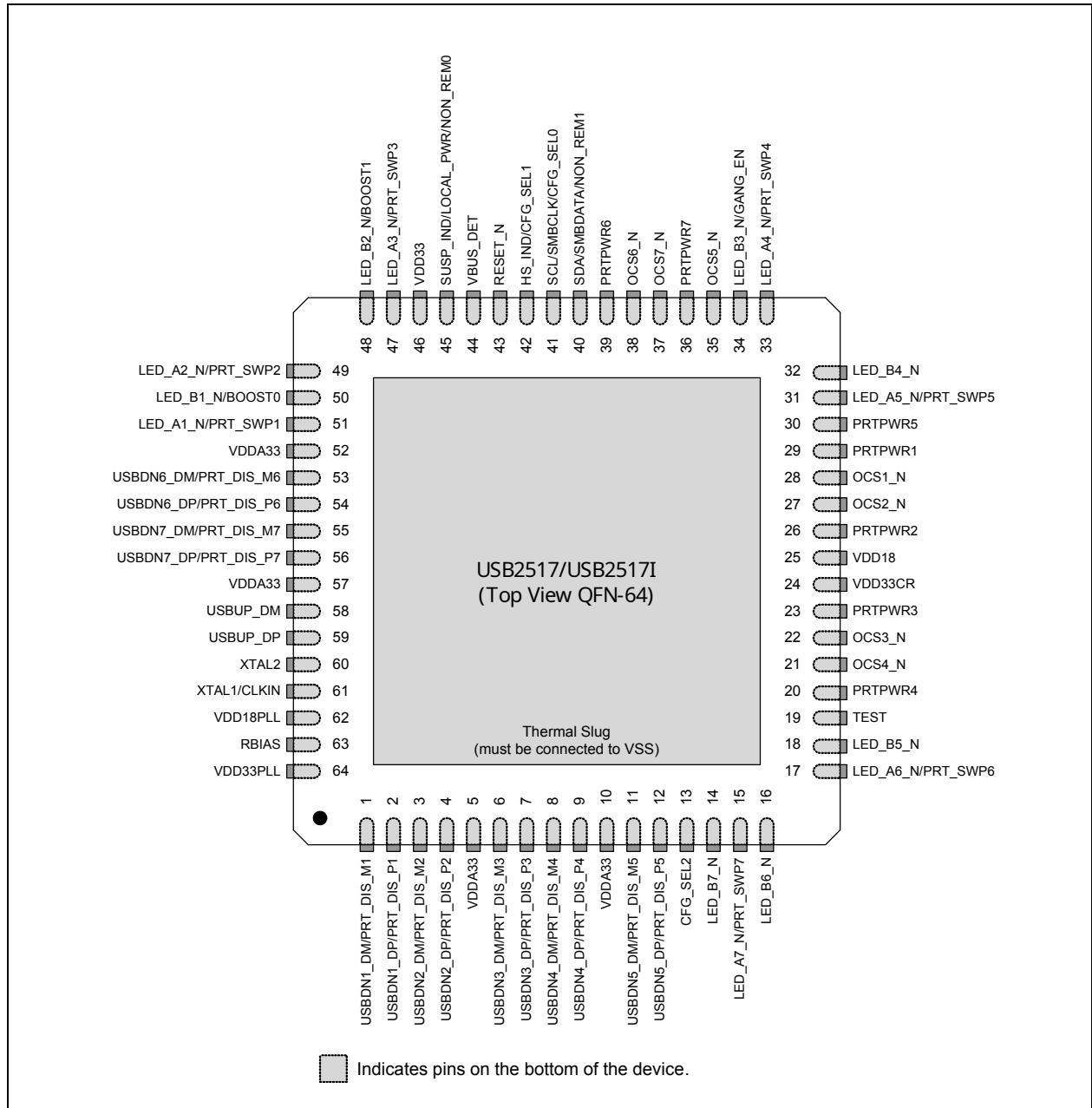


3.0 PIN CONFIGURATION

FIGURE 3-1: USB2517I 64-PIN QFN DIAGRAM



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4.0 PIN TABLE

4.1 64-Pin List

TABLE 4-1: USB2517I 64-PIN TABLE

Upstream USB 2.0 Interfaces (3 pins)			
USBUP_DP	USBUP_DM	VBUS_DET	
Downstream 7-Port USB 2.0 Interfaces (43 Pins)			
USBDN1_DP/ PRT_DIS_P1	USBDN2_DP/ PRT_DIS_P2	USBDN3_DP/ PRT_DIS_P3	USBDN4_DP/ PRT_DIS_P4
USBDN5_DP/ PRT_DIS_P5	USBDN6_DP/ PRT_DIS_P6	USBDN7_DP/ PRT_DIS_P7	USBDN1_DM/ PRT_DIS_M1
USBDN2_DM/ PRT_DIS_M2	USBDN3_DM/ PRT_DIS_M3	USBDN4_DM/ PRT_DIS_M4	USBDN5_DM/ PRT_DIS_M5
USBDN6_DM/ PRT_DIS_M6	USBDN7_DM/ PRT_DIS_M7	LED_A1_N/ PRT_SWP1	LED_A2_N/ PRT_SWP2
LED_A3_N/ PRT_SWP3	LED_A4_N/ PRT_SWP4	LED_A5_N/ PRT_SWP5	LED_A6_N/ PRT_SWP6
LED_A7_N/ PRT_SWP7	LED_B1_N/ BOOST0	LED_B2_N/ BOOST1	LED_B3_N/ GANG_EN
LED_B4_N	LED_B5_N	LED_B6_N	LED_B7_N
PRT_PWR1	PRT_PWR2	PRT_PWR3	PRT_PWR4
PRT_PWR5	PRT_PWR6	PRT_PWR7	OCS1_N
OCS2_N	OCS3_N	OCS4_N	OCS5_N
OCS6_N	OCS7_N	RBIAS	
Serial Port Interface (4 Pins)			
SDA/ SMBDATA/ NON_REM1	SCL/ SMBCLK/ CFG_SEL0	HS_IND/ CFG_SEL1	CFG_SEL2
MISC (5 Pins)			
XTAL1/CLKIN	XTAL2	SUSP_IND/ LOCAL_PWR/ NON_REM0	RESET_N
TEST			

TABLE 4-1: USB2517I 64-PIN TABLE (CONTINUED)

Analog Power (6 Pins)			
VDD18PLL	VDD33PLL	(4) VDDA33	
Digital Power, Ground (3 Pins)			
VDD33	VDD18	VDD33CR	
Total 64			

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5.0 PIN DESCRIPTIONS AND BUFFER TYPE DESCRIPTIONS

5.1 Pin Descriptions

This section provides a detailed description of each signal. The signals are arranged in functional groups according to their associated interface.

The “N” symbol in the signal name indicates that the active, or asserted, state occurs when the signal is at a low voltage level. When “N” is not present before the signal name, the signal is asserted when at the high voltage level.

The terms assertion and negation are used exclusively. This is done to avoid confusion when working with a mixture of “active low” and “active high” signals. The term assert, or assertion, indicates that a signal is active, independent of whether that level is represented by a high or low voltage. The term negate, or negation, indicates that a signal is inactive.

TABLE 5-1: USB2517/USB2517I PIN DESCRIPTIONS

Symbol	64 QFN	Buffer Type	Description
Upstream USB Interfaces			
USBUP_DP USBUP_DM	59 58	IO-U	USB Bus Data These pins connect to the upstream USB bus data signals (Host port, or upstream hub).
VBUS_DET	44	I/O12	Detect Upstream VBUS Power Detects state of Upstream VBUS power. The MCHP Hub monitors VBUS_DET to determine when to assert the internal D+ pull-up resistor (signaling a connect event). When designing a detachable hub, this pin must be connected to the VBUS power pin of the USB port that is upstream from the hub. For self-powered applications with a permanently attached host, this pin must be connected to 3.3V or 5.0V (typically VDD33).
Downstream 7-Port USB 2.0 Interfaces			
USBDN[7:1]_DP/ PRT_DIS_P[7:1] & USBDN[7:1]_DM/ PRT_DIS_M[7:1]	56 54 12 9 7 4 2 55 53 11 8 6 3 1	IO-U	Hi-Speed USB Data & Port Disable Strap Option USBDN_DP[7:1] / USBDN_DM[7:1]: These pins connect to the downstream USB peripheral devices attached to the hub's port. Downstream Port Disable Strap option: PRT_DIS_P[7:1] / PRT_DIS_M[7:1]: If the strap is enabled by package and configuration settings (see Table 5-2), this pin will be sampled at RESET_N negation to determine if the port is disabled. To disable, pull up with 10K resistor to 3.3V.
PRT_PWR[7:1]	36 39 30 20 23 26 29	O12	USB Power Enable Enables power to USB peripheral devices downstream. Note: The hub supports active high power controllers only!

TABLE 5-1: USB2517/USB2517I PIN DESCRIPTIONS (CONTINUED)

Symbol	64 QFN	Buffer Type	Description
LED_A[7:1]_N/ PRT_SWP[7:1]	15 17 31 33 47 49 51	I/O12	Port LED Indicators & Port Swap strapping option Indicator LED for ports 1-7. Will be active low when LED support is enabled via EEPROM or SMBus. If this strap is enabled by package and configuration settings (see Table 5-2), this pin will be sampled at RESET_N negation to determine the electrical connection polarity of the downstream USB Port pins (USB_DP and USB_DM). Also, the active state of the LED will be determined as follows: '0' = Port Polarity is normal, LED is active high. '1' = Port Polarity (USB_DP and USB_DM) is swapped, LED is active low.
LED_B[7:4]_N	14 16 18 32	I/O12	Enhanced Indicator Port 4-7 LED Enhanced Indicator LED for ports 4-7. Will be active low when LED support is enabled via EEPROM or SMBus.
LED_B3_N/ GANG_EN	34	I/O12	Enhanced Port 3 LED, Gang Power, and Over-current Strap Option Enhanced Indicator LED for port 3. Will be active low when LED support is enabled via EEPROM or SMBus. GANG_EN: Selects between Gang or Individual Port power and Over-current sensing. If this strap is enabled by package and configuration settings (see Table 5-2), this pin will be sampled at RESET_N negation to determine the mode as follows: '0' = Individual sensing & switching, and LED_B3_N is active high. '1' = Ganged sensing & switching, and LED_B3_N is active low.

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TABLE 5-1: USB2517/USB2517I PIN DESCRIPTIONS (CONTINUED)

Symbol	64 QFN	Buffer Type	Description
LED_B[2:1]_N/ BOOST[1:0]	48 50	I/O12	Enhanced Port [2:1] LED & PHY Boost strapping option Enhanced Indicator LED for ports 1 & 2. Will be active low when LED support is enabled via EEPROM or SMBus. BOOST[1:0]: If this strap is enabled by package and configuration settings (see Table 5-2), this pin will be sampled at RESET_N negation to determine if all PHY ports (upstream and downstream) operate at a normal or boosted electrical level. Also, the active state of the LEDs will be determined as follows: See Section 7.2.1.26, "Register F6h: Boost_Up" and Section 7.2.1.28, "Register F8h: Boost_4:0". BOOST[1:0] = BOOST_IOUT[1:0] BOOST[1:0] = '00', LED_B2_N is active high, LED_B1_N is active high. BOOST[1:0] = '01', LED_B2_N is active high, LED_B1_N is active low. BOOST[1:0] = '10', LED_B2_N is active low, LED_B1_N is active high. BOOST[1:0] = '11', LED_B2_N is active low, LED_B1_N is active low.
OCS[7:1]_N	37 38 35 21 22 27 28	IPU	Over-current Sense Input from external current monitor indicating an over-current condition. {Note: Contains internal pull-up to 3.3V supply}
RBIAS	63	I-R	USB Transceiver Bias A 12.0kΩ (+/- 1%) resistor is attached from the ground to this pin to set the transceiver's internal bias settings.

TABLE 5-1: USB2517/USB2517I PIN DESCRIPTIONS (CONTINUED)

Symbol	64 QFN	Buffer Type	Description
Serial Port Interface			
SDA/ SMBDATA/ NON_REM1	40	I/OSD12	Serial Data / SMB Data
			NON_REM1: Non-removable port strap option. If this strap is enabled by package and configuration settings (see Table 5-2) this pin will be sampled (in conjunction with SUSP_IND/LOCAL_PWR/NON_REM0) at RESET_N negation to determine if ports [3:1] contain permanently attached (non-removable) devices: NON_REM[1:0] = '00', All ports are removable, NON_REM[1:0] = '01', Port 1 is non-removable, NON_REM[1:0] = '10', Ports 1 & 2 are non-removable, NON_REM[1:0] = '11', Ports 1, 2 & 3 are non-removable.
SCL/ SMBCLK/ CFG_SEL0	41	I/OSD12	Serial Clock (SCL)
			SMBus Clock (SMBCLK) Configuration Select_SEL0: The logic state of this multifunctional pin is internally latched on the rising edge of RESET_N (RESET_N negation), and will determine the hub configuration method as described in Table 5-2, "USB2517I SMBUS or EEPROM Interface Behavior".
HS_IND/ CFG_SEL1	42	I/O12	Hi-Speed Upstream port indicator & Configuration Programming Select
			HS_IND: High Speed Indicator for upstream port connection speed. The active state of the LED will be determined as follows: CFG_SEL1 = '0', HS_IND is active high, CFG_SEL1 = '1', HS_IND is active low, 'Asserted' = Hub is connected at HS 'Negated' = Hub is connected at FS CFG_SEL1: The logic state of this pin is internally latched on the rising edge of RESET_N (RESET_N negation), and will determine the hub configuration method as described in Table 5-2, "USB2517I SMBUS or EEPROM Interface Behavior".
CFG_SEL2	13	I	Configuration Programming Select This pin is not available in all packages; it is held to a logic '0' when not available. The logic state of this pin is internally latched on the rising edge of RESET_N (RESET_N negation), and will determine the hub configuration method as described in Table 5-2, "USB2517I SMBUS or EEPROM Interface Behavior".

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TABLE 5-1: USB2517/USB2517I PIN DESCRIPTIONS (CONTINUED)

Symbol	64 QFN	Buffer Type	Description
MISC			
XTAL1/ CLKIN	61	ICLKx	Crystal Input/External Clock Input 24MHz crystal or external clock input. This pin connects to either one terminal of the crystal or to an external 24MHz clock when a crystal is not used.
XTAL2	60	OCLKx	Crystal Output 24MHz Crystal This is the other terminal of the crystal. It can be treated as a no connect when an external clock source is used to drive XTAL1/CLKIN. This output must not be used to drive any external circuitry other than the crystal circuit.
RESET_N	43	IS	RESET Input The system can reset the chip by driving this input low. The minimum active low pulse is 1 μ s. When the RESET_N pin is pulled to VDD33, the internal POR (Power on Reset) is enabled and no external reset circuitry is required. The internal POR holds the internal logic in reset until the power supplies are stable.
SUSP_IND/ LOCAL_PWR/ NON_REM0	45	I/O12	Active/Suspend status LED or Local-Power & Non-Removable Strap Option Suspend Indicator: Indicates the USB state of the hub. 'negated' = Unconfigured or configured and in USB suspend 'asserted' = Hub is configured, and is active (i.e., not in suspend) Local Power: Detects availability of local self-power source. Low = Self/local power source is NOT available (i.e., Hub gets all power from the upstream USB VBus). High = Self/local power source is available. NON_REM0 Strap Option: If this strap is enabled by package and configuration settings (see Table 5-2, "USB2517I SMBUS or EEPROM Interface Behavior"), this pin will be sampled (in conjunction with NON_REM1) at RESET_N negation to determine if ports [3:1] contain permanently attached (non-removable) devices. Also, the active state of the LED will be determined as follows: NON_REM[1:0] = '00', All ports are removable, and the LED is active high NON_REM[1:0] = '01', Port 1 is non-removable, and the LED is active low NON_REM[1:0] = '10', Ports 1 & 2 are non-removable, and the LED is active high NON_REM[1:0] = '11', Ports 1, 2 & 3 are non-removable, and the LED is active low

TABLE 5-1: USB2517/USB2517I PIN DESCRIPTIONS (CONTINUED)

Symbol	64 QFN	Buffer Type	Description
TEST	19	IPD	TEST pin XNOR continuity tests all signal pins on the hub. Please contact your MCHP representative for a detailed description of how this test mode is enabled and utilized.
Power, Ground, No Connect			
VDD18	25		VDD Core +1.8V core power. This pin must have a 1.0 μ F (or greater) $\pm$ 20% (ESR <0.1 Ω) capacitor to VSS.
VDD33PLL	64		VDD 3.3 PLL Regulator Reference +3.3V power supply for the Digital I/O. If the internal PLL 1.8V regulator is enabled, then this pin acts as the regulator input.
VDD18PLL	62		VDD PLL +1.8V Filtered analog power for internal PLL. This pin must have a 1.0 μ F (or greater) $\pm$ 20% (ESR <0.1 Ω) capacitor to VSS.
VDD33	46		VDD I/O +3.3V Digital I/O power
VDDA33	5 10 52 57		VDD Analog I/O +3.3V Filtered analog PHY power which is shared between adjacent ports.
VDD33CR	24		VDDIO/VDD 3.3 Core Regulator Reference +3.3V power supply for the Digital I/O. If the internal core regulator is enabled, then VDD33CR acts as the regulator input.
Ground	VSS	Slug	Ground

TABLE 5-2: USB2517I SMBUS OR EEPROM INTERFACE BEHAVIOR

CFG_SEL2	CFG_SEL1	CFG_SEL0	SMBus or EEPROM Interface Behavior
0	0	0	Internal Default Configuration • Strap Options are Enabled
0	0	1	Configured as an SMBus slave for external download of user-defined descriptors • SMBus slave address is '0101100' • Strap Options are Disabled • All Settings are Controlled by Registers
0	1	0	Internal Default Configuration • Strap Options are Enabled • Bus Power Operation • LED Mode = USB

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TABLE 5-2: USB2517I SMBUS OR EEPROM INTERFACE BEHAVIOR (CONTINUED)

CFG_SEL2	CFG_SEL1	CFG_SEL0	SMBus or EEPROM Interface Behavior
0	1	1	2-Wire I ² C EEPROMS are supported • Strap Options are Disabled • All Settings are Controlled by Registers
1	0	0	Internal Default Configuration • Strap Options are Disabled • Dynamic Power Switching is Enabled
1	0	1	Internal Default Configuration • Strap Options are Disabled • Dynamic Power Switching is Enabled • LED Mode = USB
1	1	0	Internal Default Configuration • Strap Options are Disabled
1	1	1	Internal Default Configuration • Strap Options are Disabled • LED Mode = USB • Ganged Power Switching • Ganged Over-Current Sensing

5.2 Buffer Type Descriptions

TABLE 5-3: USB2517/USB2517I BUFFER TYPE DESCRIPTIONS

Buffer	Description
I	Input.
IPD	Input with internal weak pull-down resistor.
IPU	Input with internal weak pull-up resistor.
IS	Input with Schmitt trigger.
O12	Output 12mA.
I/O12	Input/Output buffer with 12mA sink and 12mA source.
I/OSD12	Open drain...12mA sink with Schmitt trigger, and must meet I ² C-Bus Specification Version 2.1 requirements.
ICLKx	XTAL clock input.
OCLKx	XTAL clock output.
I-R	RBIAS.
IO-U	Analog Input/Output Defined in USB specification.

6.0 LED USAGE DESCRIPTION

6.1 LED Functionality

The hub supports 2 different (mutually exclusive) LED modes. The USB Mode provides 14 LED's that conform to the USB 2.0 specification functional requirements for Green and Amber LED's. The LED Mode "Speed indicator" provides the downstream device connection speed.

6.1.1 USB MODE 14-WIRE

The LED_A[7:1]_N pins are used to provide Green LED support as defined in the USB 2.0 specification. The LED_B[7:1]_N pins are used to provide Amber LED support as defined in the USB 2.0 specification. The USB Specification defines the LED's as port status indicators for the downstream ports. Please note that no indication of port speed is possible in this mode. The pins are utilized as follows:

- LED_A1_N = Port 1 green LED
- LED_A2_N = Port 2 green LED
- LED_A3_N = Port 3 green LED
- LED_A4_N = Port 4 green LED
- LED_A5_N = Port 5 green LED
- LED_A6_N = Port 6 green LED
- LED_A7_N = Port 7 green LED
- LED_B1_N = Port 1 amber LED
- LED_B2_N = Port 2 amber LED
- LED_B3_N = Port 3 amber LED
- LED_B4_N = Port 4 amber LED
- LED_B5_N = Port 5 amber LED
- LED_B6_N = Port 6 amber LED
- LED_B7_N = Port 7 amber LED

6.1.2 LED MODE SPEED INDICATION

The LED_A[7:1]_N pins are used to provide connection status as well as port speed by using dual color LED's. This scheme requires that the LED's be in the same package, and that a third color is produced so that the user perceives both LED's as being driven "simultaneously".

The LED_A[7:1] pins used in this mode are connected to 7 dual color LED's (each LED pair in a single package). These pins indicate the USB speed of each attached downstream device.

Each dual color LED provides two separate colors (commonly Green and Red). If each of these separate colors are pulsed on and off at a rapid rate, a user will see a third color (in this example, Orange). Using this method, 4 different "color" states are possible (Green, Red, Orange, and Off).

FIGURE 6-1: DUAL COLOR LED IMPLEMENTATION EXAMPLE

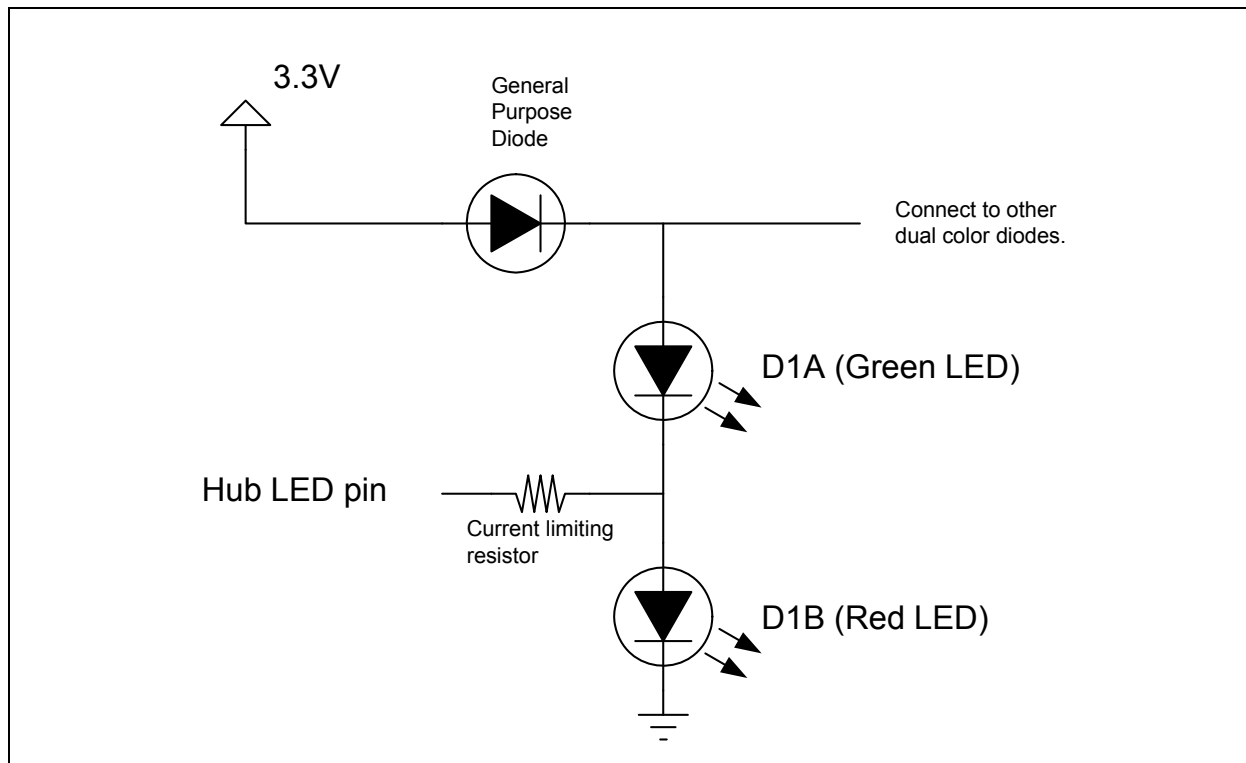


Figure 6-1 shows a simple example of how this LED circuit will be implemented. The circuit should be replicated for each of the 7 LED pins on the HUB. In this circuit, when the LED pin is driven to a logic low state, the Green LED will light up. When the LED pin is driven to a Logic High state the Red LED will light up. When a 1 KHz square wave is driven out on the LED pin, the Green and Red LED's will both alternately light up giving the effect of the color Orange. When nothing is driven out on the LED pin (i.e. the pin floats to a "tri-state" condition), neither the Green nor Red LED will light up, this is the "Off" state.

The assignment is as follows:

- LED_A1_N = LED D1 (Downstream Port 1)
- LED_A2_N = LED D2 (Downstream Port 2)
- LED_A3_N = LED D3 (Downstream Port 3)
- LED_A4_N = LED D4 (Downstream Port 4)
- LED_A5_N = LED D5 (Downstream Port 5)
- LED_A6_N = LED D6 (Downstream Port 6)
- LED_A7_N = LED D7 (Downstream Port 7)

The usage is as follows:

- LED_Ax_N Driven to Logic Low = LS device attached (Green LED)
- LED_Ax_N Driven to Logic High = FS device attached (Red LED)
- LED_Ax_N Pulsed @ 1 KHz= HS device attached (Orange color by pulsing Red & Green).
- LED_Ax_N is tri-state= No devices are attached, or the hub is in suspend, LED's are off.